



Malabsorption & Diseases of Small Bowel

Intended learning outcomes:

- By the end of this lecture the student will be able to:
 1. Do a proper approach to patient with malabsorption .
 2. Manage a patient with celiac disease.

Malabsorption

Diarrhoea and weight loss in patients with a normal diet are likely to be caused by malabsorption.

Presentations

A few patients have apparently normal bowel habit, but diarrhoea usually occurs and may be watery and voluminous.

Bulky, pale and offensive stools that float in the toilet (steatorrhoea) signify fat malabsorption.

Abdominal distension, borborygmi, cramps, weight loss and undigested food in the stool may be present.

Some patients complain only of malaise and lethargy.

In others, symptoms related to deficiencies of specific vitamins, trace elements and minerals may occur.

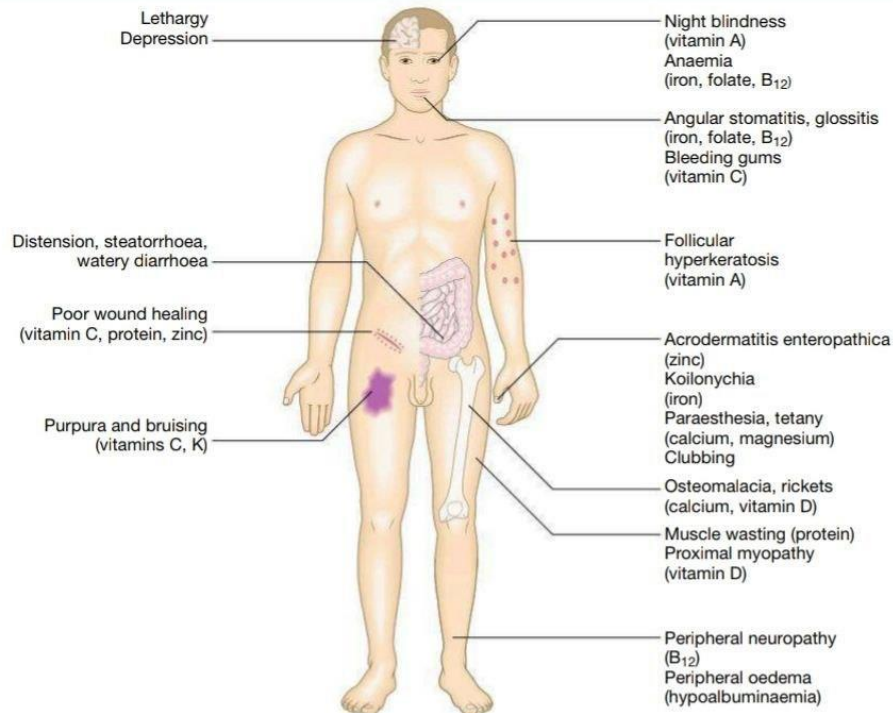


Fig. 23.22 Possible physical consequences of malabsorption.

Investigations of Malabsorption

Investigations should be performed both to confirm the presence of malabsorption and to determine the underlying cause.

Routine blood tests should be done to show the consequences of malabsorption.

Tests to exclude fat & protein malabsorption .

Other tests according to the clinical suspicion

i	23.20 Routine blood test abnormalities in malabsorption
Haematology	
•	
Microcytic anaemia (iron deficiency)	
•	
Macrocytic anaemia (folate or B ₁₂ deficiency)	
•	
Increased prothrombin time (vitamin K deficiency)	
Biochemistry	
•	
Hypoalbuminaemia	
•	
Hypocalcaemia	
•	
Hypomagnesaemia	
•	
Hypophosphataemia	
•	
Low serum zinc	

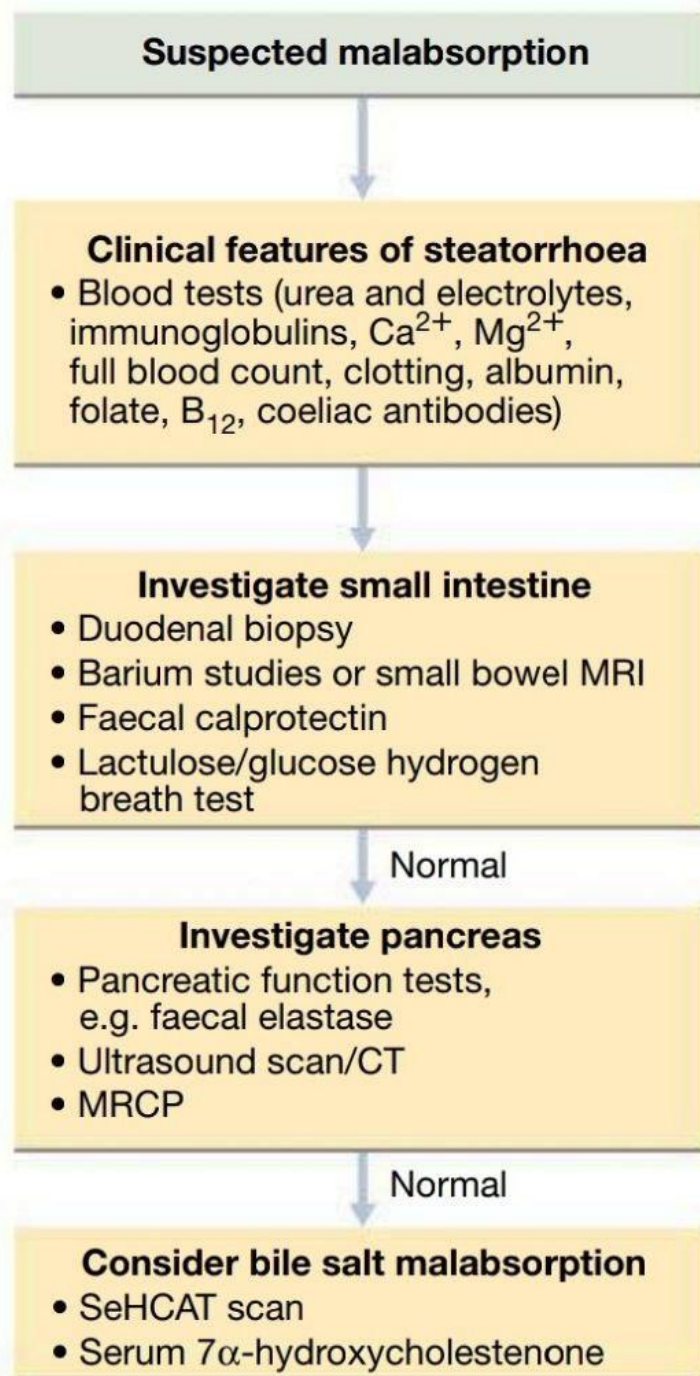


Fig. 23.23 Investigation for suspected malabsorption. (CT = computed tomography; MRCP = magnetic resonance cholangiopancreatography; MRI = magnetic resonance imaging; $^{75}\text{SeHCAT}$ = ^{75}Se -homocholic acid taurine)



Disorders causing malabsorption:

Coeliac Disease

It is an **inflammatory** disorder of the small bowel occurring in **genetically susceptible** individuals, which results from intolerance to wheat gluten and similar proteins found in rye, barley and, to a lesser extent, oats.



It can result in malabsorption and responds to a gluten-free diet. The condition occurs worldwide, but is more common in northern Europe.

It could be asymptomatic ! (50%)

These include both undiagnosed '**silent**' cases of the disease and cases of '**latent**' coeliac disease – genetically susceptible people who may later develop clinical coeliac disease.

Pathophysiology

Immunological responses to gluten play a key role.

There is a strong genetic component.

There is a strong association with human leukocyte antigen (**HLA**)-**DQ2/DQ8**.

Dysbiosis of the intestinal microbiota has been Identified.



Clinical Presentation

Coeliac disease can present at any age!

In infancy, it occurs after weaning on to cereals and typically presents with **diarrhoea**, **malabsorption** and **failure to thrive**.

In older children, it may present with non-specific features, such as **delayed growth**.

Features of malnutrition are found on examination and mild abdominal distension may be present.

Affected children have growth and pubertal delay, leading to short stature in adulthood.



In **adults**, the disease usually presents during the third or fourth decade and females are affected twice as often as male.

Some have florid malabsorption, while others develop non-specific symptoms, such as tiredness, weight loss, folate deficiency or **iron deficiency anaemia** (suspect celiac in any unexplained IDA !!).

Other symptoms include oral ulceration, dyspepsia and bloating; such non-specific symptomatology is more common in older people Clinical presentation – cont.



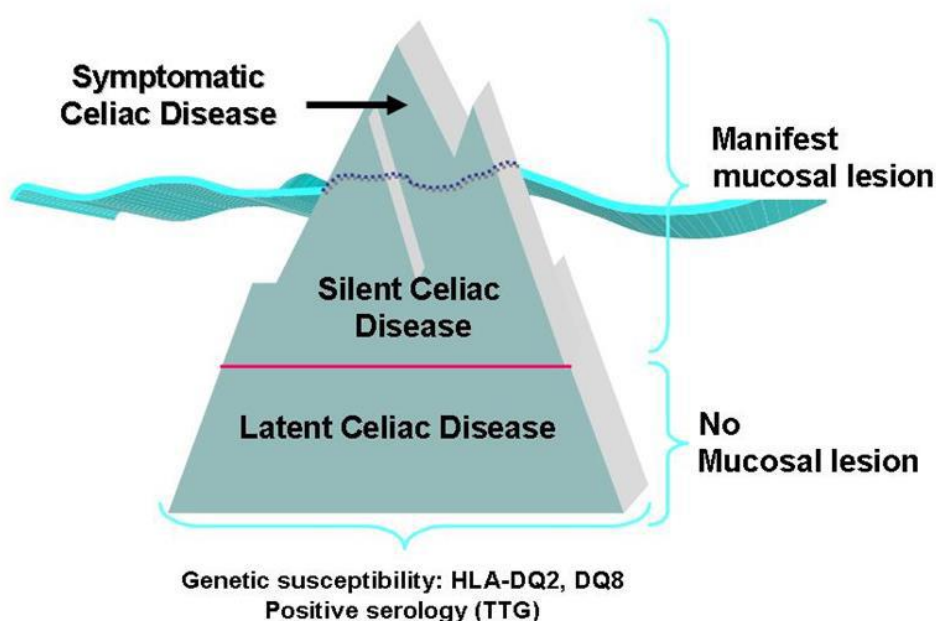
Unrecognised coeliac disease is associated with mild under-nutrition and osteoporosis.

Coeliac disease is associated with other HLA-linked autoimmune disorders and with certain other diseases !

22.49 Disease associations of coeliac disease	
• Insulin-dependent diabetes mellitus (2–8%) • Thyroid disease (5%) • Primary biliary cirrhosis (3%) • Sjögren's syndrome (3%) • IgA deficiency (2%) • Pernicious anaemia • Sarcoidosis • Neurological complications: encephalopathy, cerebellar atrophy, peripheral neuropathy, epilepsy	• Myasthenia gravis • Dermatitis herpetiformis • Down's syndrome • Enteropathy-associated T-cell lymphoma • Small bowel carcinoma • Squamous carcinoma of oesophagus • Ulcerative jejunitis • Pancreatic insufficiency • Microscopic colitis • Splenic atrophy

Coeliac disease is an Iceberg phenomena !

The Celiac Iceberg



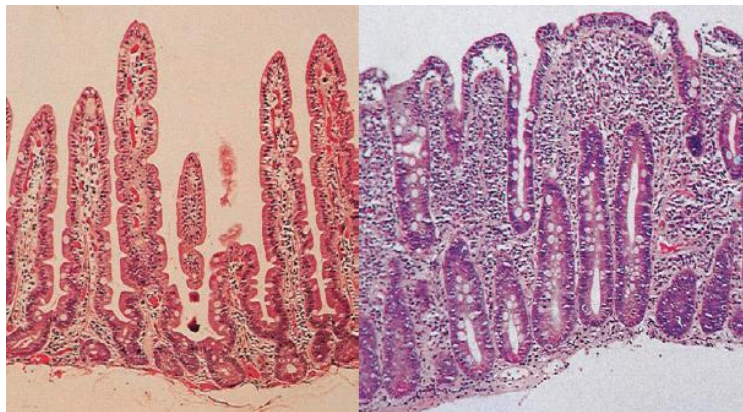


Investigations

1. **Duodenal biopsy**: endoscopic small bowel biopsy is the gold standard. Endoscopic appearances should not preclude biopsy, as the mucosa usually looks normal.

Adequate number of biopsies should be taken.

Other causes of villous atrophy should be Considered.



22.50 Important causes of subtotal villous atrophy

- | | |
|----------------------------|------------------------------|
| • Coeliac disease | • Giardiasis |
| • Tropical sprue | • Hypogammaglobulinaemia |
| • Dermatitis herpetiformis | • Radiation |
| • Lymphoma | • Whipple's disease |
| • HIV-related enteropathy | • Zollinger–Ellison syndrome |

2. **Antibodies**: anti tissue transglutaminase antibodies IgA and IgG can be used as valuable screening test , **but are not a diagnostic substitute for small bowel biopsy**.

If the antibody screen is positive, adult patients should remain on a gluten-containing diet until duodenal biopsies are taken.

Anti endomysial Ab : They are sensitive (85%– 95%) and specic (approximately 99%) for the diagnosis , but may be not reliable in cases of IgA deficiency.

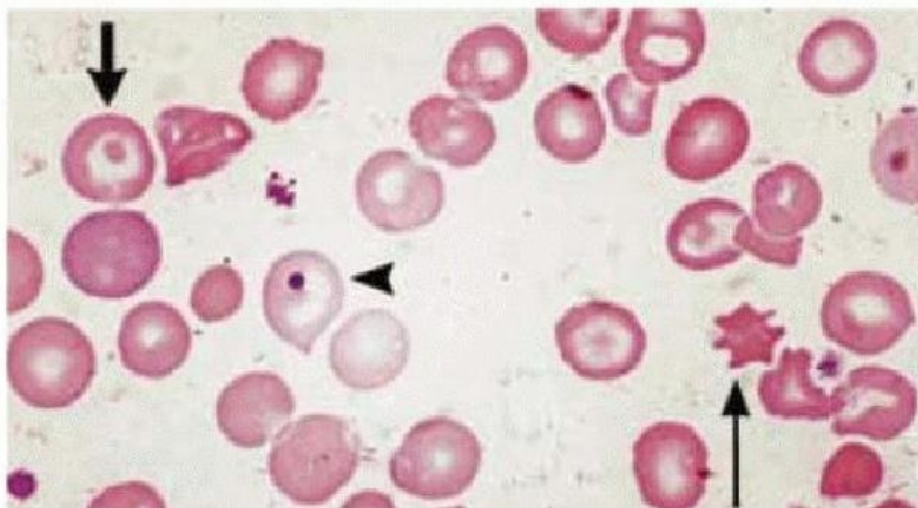


3. *Haematology and biochemistry:*

- A full blood count may show microcytic or macrocytic anaemia from iron or folate deficiency and features of hyposplenism (target cells, spherocytes and Howell–Jolly bodies).

Biochemical tests may reveal reduced concentrations of calcium, magnesium, total protein, albumin or vitamin D. Serum IgA measurement is required to ensure an appropriate IgA response.

Film in hyposplenism: target cell (short arrow), acanthocyte (long arrow) and a Howell-Jolly body (arrow head).



4. *Other tests:*

Measurement of bone density (DXA scan) should be considered to look for osteopenia or osteoporosis, especially in older patients and post-menopausal women.

Management

1. Gluten free diet : exclusion of wheat, rye, barley and initially oats
2. Frequent dietary counseling



3. Correction of vitamins , minerals deficiencies & electrolyte disturbances
4. Some patients who fail to respond adequately to a gluten-free diet may require therapy with glucocorticoids or immunosuppressive drugs, and may be classified as refractory coeliac disease (RCD).

Complications

1. Risk of malignancies:

A. enteropathy associated T-cell lymphoma

B. small bowel carcinoma

C. squamous carcinoma of the oesophagus

2. Ulcerative jejuno-ileitis: This may present with fever, pain, obstruction or perforation.

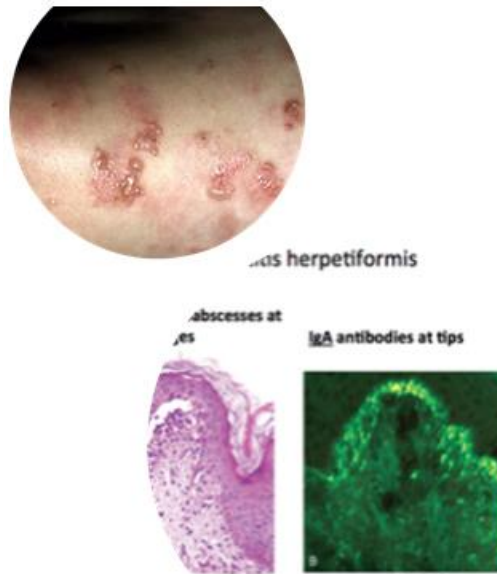
3. Osteoporosis and osteomalacia

Still Diarrhea In Spite of Gluten Free Diet?

- Non Compliance .
- Associations : pancreatic insufficiency , microscopic colitis, inflammatory bowel disease.
- Complications: ulcerative jejunitis , enteropathy associated T-cell lymphoma, small bowel carcinoma and refractory coeliac disease.

Dermatitis Herpetiformis

- Intensely itchy blisters over the elbows, knees, back and buttocks.
- Immunofluorescence shows granular or linear IgA deposition at the dermo-epidermal junction



Almost all patients have partial villous atrophy on duodenal biopsy, identical to that seen in coeliac disease, even though they usually have no gastrointestinal symptoms.

Fewer than 10% of coeliac patients have evidence of dermatitis herpetiformis.

The rash usually responds to a **gluten free diet** but some patients require additional treatment with dapsone.

Tropical Sprue

Chronic, progressive malabsorption in a patient in or from a tropical country, associated with abnormalities of small intestinal structure and function.

The disease occurs mainly in the West Indies and in southern India, Malaysia and Indonesia.

The condition often begins after an acute diarrhoeal illness. Small bowel bacterial overgrowth with *Escherichia coli*, *Enterobacter* and *Klebsiella* is frequently seen.

The changes closely resemble those of coeliac disease.

Clinically : features of malabsorption.



Management : Tetracycline (250 mg 4 times daily for 28 days) is the treatment of choice and brings about long-term remission or cure. In most patients, pharmacological doses of folic acid (5 mg daily) improve symptoms and jejunal Morphology .

To summarize:

- Malabsorption can cause watery diarrhoea or steatorrhoea.
- Celiac disease is a common cause of malabsorption.
- Atypical presentation is common in celiac disease
- The gold standard test for celiac is duodenal biopsy.
- Treatment of celiac disease is gluten free diet.

References:

Davidson's principles and Practice of medicine, 24rd edition , 2023.

لا تنسوا أهلنا في فلسطين و السودان من صادق دعائكم